

ITAO7102 Human Resources Analytics

Assignment

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Part 1: Visual Analytics and Storytelling Using Tableau

Introduction to Part 1

This section presents a visual analytics storyboard developed using Tableau to support data-driven decision-making within a large pharmaceutical organisation experiencing high levels of employee attrition. The organisation currently relies on static Excel-based reporting, which limits its ability to explore workforce trends, identify emerging risks, and communicate insights effectively to senior stakeholders. Tableau was therefore used to transform Human Resource Information System (HRIS) data into an interactive and structured narrative that highlights workforce context, identifies sources of attrition-related tension, and proposes evidence-based resolutions.

The storyboard follows a classical narrative structure comprising **Context**, **Tension**, and **Resolution**, allowing decision-makers to move logically from understanding the workforce composition to identifying attrition drivers and, finally, to exploring actionable HR interventions.

Context: Understanding the Workforce Landscape

The first part of the storyboard focuses on establishing organisational context by providing an overview of the workforce and key employee characteristics. This is a critical step in people analytics, as attrition cannot be meaningfully interpreted without first understanding who the employees are and how the workforce is structured.

The *Workforce Overview* dashboard presents the distribution of employees across departments, job roles, job levels, and working arrangements (office-based or hybrid). Demographic variables such as age, gender, education level, and marital status are also visualised to provide a holistic picture of the organisation's human capital. These visualisations allow stakeholders to identify workforce concentration areas and structural characteristics that may later be associated with higher attrition risk.

By consolidating multiple workforce indicators into a single dashboard, Tableau enables rapid sense-making that would be difficult to achieve using traditional Excel reports. Interactive filtering further allows managers to explore specific segments of the workforce, such as departments or job levels, enhancing managerial insight and engagement.

Overall, this dashboard establishes a clear baseline by answering the fundamental question: *“Who makes up our workforce, and how is it organised?”* This contextual understanding sets the foundation for exploring attrition-related challenges in subsequent sections of the storyboard.

Tension: Identifying Attrition Patterns and Risk Areas

The second section of the storyboard introduces the central business challenge: employee attrition. This stage focuses on uncovering where attrition is occurring, which groups are most affected, and which employment conditions are associated with higher turnover rates.

The *Attrition Patterns and Emerging Risk Areas* dashboard visualises attrition rates by department, job role, overtime status, work–life balance, and job satisfaction. These visualisations reveal that attrition is not evenly distributed across the organisation. Certain departments exhibit disproportionately high attrition rates, suggesting potential role-specific or workload-related pressures. Similarly, employees reporting poor work–life balance and lower job satisfaction demonstrate significantly higher likelihoods of leaving the organisation.

Overtime emerges as a particularly important factor, with employees who regularly work overtime showing elevated attrition rates. This finding suggests that excessive workload and insufficient recovery time may be contributing to employee disengagement and eventual turnover. These insights align with established HR literature, which identifies workload stress and work–life imbalance as key predictors of voluntary attrition.

By visually juxtaposing attrition with multiple employee characteristics, the dashboard makes the tension within the organisation both visible and actionable. The use of colour encoding and comparative charts highlights high-risk groups clearly, enabling decision-makers to prioritise attention and resources. Without such visual analytics, these patterns may remain hidden within static tabular reports.

This section addresses the critical question: *“Where is attrition coming from, and why does it represent a risk to the organisation?”* If left unaddressed, these patterns may lead to increased recruitment costs, loss of organisational knowledge, and reduced operational performance.

Resolution: From Insight to Action

The final section of the storyboard focuses on resolution by translating analytical insights into actionable HR interventions. The *Key Drivers of Attrition and Strategic HR Interventions* dashboard synthesises findings from earlier sections and highlights the most influential levers available to management.

Visualisations in this dashboard examine the relationship between attrition, overtime, and compensation levels. The analysis indicates that employees experiencing high workloads without corresponding compensation adjustments are more likely to leave. This insight suggests that both workload management and pay equity should be considered as part of a broader retention strategy.

By allowing users to filter by department or job role, the dashboard supports targeted interventions rather than generic, organisation-wide policies. For example, departments with persistently high overtime-related attrition may benefit from workforce planning adjustments, additional staffing, or revised performance expectations. Similarly, compensation reviews may be prioritised for roles where attrition is high despite strong performance outcomes.

The resolution stage demonstrates how Tableau supports not only descriptive analytics but also strategic decision-making. Rather than merely reporting attrition figures, the storyboard enables HR leaders to explore “what-if” scenarios and identify practical actions that can be implemented in the short, medium, and long term.

This section answers the final narrative question: “*What should the organisation do next?*” The storyboard therefore closes the loop between data, insight, and action.

Value of Tableau for HR Decision-Making

Across all three stages of the storyboard, Tableau demonstrates clear advantages over traditional Excel-based reporting. Interactive dashboards facilitate exploration, pattern recognition, and stakeholder engagement, while the storyboard structure ensures analytical coherence and narrative clarity. By integrating multiple HR metrics into a single visual framework, Tableau supports evidence-based decision-making and enhances the strategic role of HR analytics within the organisation.

Summary

In summary, the Tableau storyboard successfully uncovers a data-driven story of employee attrition within the pharmaceutical organisation. By structuring the analysis around **Context**, **Tension**, and **Resolution**, the storyboard enables stakeholders to understand workforce composition, identify attrition risk factors, and explore targeted HR interventions. This approach illustrates how advanced dashboarding tools can transform HR data into strategic insight, supporting more informed and proactive people management decisions.

Part 2: Predictive Analytics for Employee Attrition

Introduction

Employee attrition represents a significant strategic challenge for organisations, particularly within knowledge-intensive industries such as pharmaceuticals, where employee expertise, continuity, and tacit knowledge are critical to operational performance. High levels of voluntary turnover are associated with increased recruitment and training costs, disruption to project delivery, and reduced organisational productivity. As a result, the ability to anticipate attrition risk and intervene proactively has become an important application of Human Resource (HR) analytics (Fitz-enz & Mattox, 2014).

The objective of this analysis is to develop a predictive model capable of identifying employees who are more likely to leave the organisation voluntarily. Using Human Resource Information System (HRIS) data provided for the organisation, predictive analytics techniques were applied to move beyond descriptive analysis and towards evidence-based decision-making. Specifically, a classification model was developed using KNIME to predict employee attrition and to identify the key factors associated with voluntary turnover. The insights derived from this model are intended to support senior management in designing targeted retention strategies and improving overall workforce stability.

This approach is consistent with existing HR analytics literature, which emphasises the role of predictive modelling in enabling more informed, proactive, and strategic people management decisions, rather than relying on reactive reporting after attrition has already occurred (Provost & Fawcett, 2013).

Data Understanding and Preparation

Data Understanding and Quality Assessment

The dataset used in this analysis was extracted from the organisation's HRIS and contains individual-level employee data covering demographic characteristics, job-related attributes, compensation variables, engagement indicators, and attrition outcomes. The target variable for the predictive model was **Attrition**, defined as a binary outcome indicating whether an employee voluntarily left the organisation.

Initial exploration revealed a combination of numerical and categorical variables, which is typical for HR datasets. Continuous variables included age, years with the company, monthly income, and training time, while categorical variables included department, job role, overtime status, and work-life balance. The exploratory analysis also identified the presence of missing values in several variables, as well as columns that were not appropriate for predictive modelling, such as unique employee identifiers.

From a data quality perspective, the dataset was largely complete and internally consistent. However, appropriate preprocessing was required to ensure the data was suitable for model development, validation, and interpretation.

Data Preparation and Preprocessing

Data preparation was conducted in KNIME using a structured and transparent workflow to ensure analytical rigour and reproducibility.

First, the dataset was imported using the **Excel Reader** node. A **Column Filter** node was then applied to remove variables that did not provide predictive value. Specifically, employee identifiers and date-of-birth fields were excluded, as these variables do not explain employee behaviour and could introduce noise or bias into the model. The retained variables were selected based on their relevance to employee characteristics, job conditions, compensation, and engagement, in line with established HR analytics practice (Kuhn & Johnson, 2013).

Missing data were addressed using the **Missing Value** node. For numerical variables, missing values were imputed using the median, as this approach is robust to outliers and preserves the underlying distribution of the data. For categorical variables, missing values were replaced using the most frequent category. This strategy minimised information loss while maintaining dataset integrity.

To support robust model validation, the dataset was partitioned using the **X-Partitioner** node. Cross-validation was selected rather than a single train–test split to reduce the risk of overfitting and to ensure that model performance metrics reflected the entire dataset. This approach is widely recommended in predictive modelling, particularly when working with moderate-sized datasets (Kuhn & Johnson, 2013).

Model Building

Model Selection and Configuration

A **Decision Tree** classification model was selected for this analysis. Decision trees are commonly used in HR analytics because they provide transparent, interpretable results and generate intuitive decision rules that can be easily communicated to non-technical stakeholders. Unlike complex “black box” models, decision trees allow HR managers to understand how specific employee attributes contribute to attrition risk.

The **Decision Tree Learner** node was connected to the training partitions generated by the X-Partitioner, with **Attrition** specified as the target (class) variable. The trained model was then applied to the test partitions using the **Decision Tree Predictor** node. Predictions from each validation fold were combined using the **X-Aggregator**, ensuring that performance evaluation reflected results across all partitions. Finally, the **Scorer** node was used to assess predictive performance.

Model Performance and Evaluation

The decision tree model demonstrated strong predictive performance. The model correctly classified **1,232 employees**, while **218 employees** were misclassified. This resulted in an overall classification accuracy of approximately **85%**, indicating that the model performs substantially better than random classification.

In addition to accuracy, **Cohen’s Kappa** statistic was examined to assess agreement beyond chance. The model achieved a Cohen’s Kappa value of approximately **0.51**, indicating a moderate level of agreement. This is particularly important in attrition modelling, as HR datasets often exhibit class imbalance, with fewer employees leaving than staying. Cohen’s Kappa therefore provides a more reliable assessment of model quality than accuracy alone. Overall, the combination of relatively high accuracy and moderate Kappa suggests that the

model is robust and suitable for supporting HR decision-making, while acknowledging that no predictive model is perfectly accurate.

Interpretation of Findings

Beyond predictive performance, the decision tree model provides meaningful insights into the drivers of employee attrition. Variables related to workload, particularly **overtime**, emerged as influential predictors, indicating that sustained excessive workload increases the likelihood of voluntary turnover. Engagement-related variables, including **work–life balance** and **job satisfaction**, were also strongly associated with attrition risk.

Compensation-related variables, such as **monthly income** and **salary increases**, were linked to retention outcomes, suggesting that perceived inequities or misalignment between effort and reward may contribute to employees' decisions to leave. Additionally, tenure-related variables, particularly **years with the company**, played an important role, with higher attrition risk observed among employees in the early stages of their employment.

These findings are consistent with established HR literature, which highlights workload stress, engagement, and early-tenure experiences as critical determinants of voluntary attrition (Fitzenz & Mattox, 2014). Importantly, the interpretability of the decision tree enables these relationships to be translated into clear business rules, supporting practical application rather than purely technical insight.

Recommendations

Based on the predictive modelling results, several actionable recommendations can be made for senior management.

First, employees who regularly work overtime should be prioritised for workload review. The model indicates that sustained overtime is associated with higher attrition risk, suggesting the need for improved workforce planning, role redesign, or additional staffing in high-pressure departments.

Second, employee engagement indicators such as work–life balance and job satisfaction should be actively monitored. Targeted interventions, including manager training programmes and regular engagement surveys, may help improve employee experience and reduce voluntary turnover.

Third, early-tenure employees represent a key risk group. Structured onboarding processes, mentoring initiatives, and early career development opportunities may help address unmet expectations and improve retention among newer employees.

Finally, compensation and reward structures should be reviewed to ensure alignment with workload and performance expectations. While compensation alone may not prevent attrition, the model suggests that perceived inequities can contribute to employees' decisions to leave.

These recommendations are directly informed by the predictive analytics and support a proactive, data-driven approach to people management.

Conclusion

This analysis demonstrates how predictive analytics can be effectively applied to HRIS data to support strategic workforce decisions. By developing and evaluating a decision tree model in KNIME, the organisation is able to identify employees at higher risk of attrition and understand the factors contributing to voluntary turnover.

The combination of robust model performance, interpretability, and actionable insights highlights the value of integrating predictive analytics into HR decision-making processes. When used alongside visual analytics tools such as Tableau, predictive models enable organisations to move beyond reactive reporting towards more informed, proactive people management decisions.

Declaration of Artificial Intelligence Use

I declare that generative artificial intelligence tools were used in a supportive capacity during the preparation of this assignment. These tools were used to assist with structuring the report, improving clarity of academic writing, and refining grammar and coherence. All data analysis, dashboard development in Tableau, predictive modelling in KNIME, interpretation of results, and formulation of conclusions and recommendations were conducted independently by me.

The use of artificial intelligence did not replace original analysis or decision-making. All content has been critically reviewed, adapted, and validated to ensure accuracy, relevance, and compliance with academic integrity requirements. This assignment represents my own work and understanding of the subject matter.

References (Harvard Style)

Fitz-enz, J. and Mattox, J.R. (2014) *Predictive Analytics for Human Resources*. Hoboken, NJ: Wiley.

Kuhn, M. and Johnson, K. (2013) *Applied Predictive Modeling*. New York: Springer.

Provost, F. and Fawcett, T. (2013) *Data Science for Business*. Sebastopol, CA: O'Reilly Media.

Appendices

Appendix A: Tableau Storyboard Structure

The Tableau storyboard consists of three dashboards organised into a single story with three slides:

- **Context Dashboard:** Workforce overview including department distribution, job levels, age groups, and demographic composition.
- **Tension Dashboard:** Attrition analysis by department, work–life balance, overtime, job satisfaction, and tenure.
- **Resolution Dashboard:** Identification of key attrition drivers and potential HR interventions focusing on workload and compensation factors.

The storyboard follows a Context–Tension–Resolution narrative structure to ensure analytical coherence and managerial relevance.

Appendix B: KNIME Predictive Modelling Workflow

The KNIME workflow was constructed using the following sequence of nodes:

1. Excel Reader
2. Column Filter
3. Missing Value
4. X-Partitioner
5. Decision Tree Learner
6. Decision Tree Predictor
7. X-Aggregator

8. Scorer

This workflow enabled structured data preprocessing, cross-validated model training, and robust evaluation of predictive performance.

Appendix C: Model Performance Metrics

The decision tree classification model produced the following performance results:

- Correctly classified instances: 1,232
- Incorrectly classified instances: 218
- Overall accuracy: approximately 85%
- Error rate: approximately 15%
- Cohen's Kappa: approximately 0.51

These metrics indicate strong predictive capability with moderate agreement beyond chance.

Appendix D: Ethical Considerations

All data used in this analysis was anonymised and provided for academic purposes. No personally identifiable information was disclosed or used outside the scope of this assignment. The analysis was conducted in accordance with ethical guidelines for data handling and academic research.